

MindGrade

Predicting Academic Performance from
Student Mental Health Indicators

International Islamic University Malaysia (IIUM) — Survey Data, July 2020

[Machine Learning](#) | [Mental Health](#) | [Academic Performance](#) | [CGPA Prediction](#)

Report generated: May 21, 2026

1. Project Overview

This report presents a statistical and machine learning analysis of the relationship between student mental health and academic CGPA at IIUM, Malaysia. The dataset contains responses from 101 students covering self-reported depression, anxiety, and panic attack status, alongside demographic and academic information.

The goal is to train classifiers to predict a student's CGPA band (5 classes: 0-1.99 to 3.50-4.00) from mental health and demographic features, and to identify which factors most influence academic performance.

2. Dataset Summary

101	34%	33%
Total respondents	Report depression	Report anxiety
32%	5%	3.50 - 4.00
Report panic attacks	Sought treatment	Most common CGPA band

3. Exploratory Data Analysis

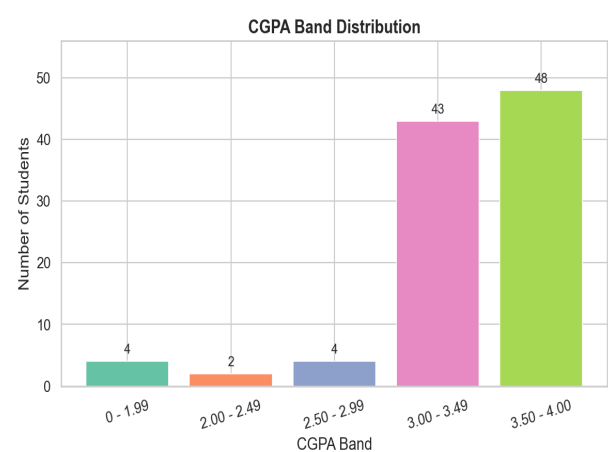


Figure 1: CGPA Band Distribution

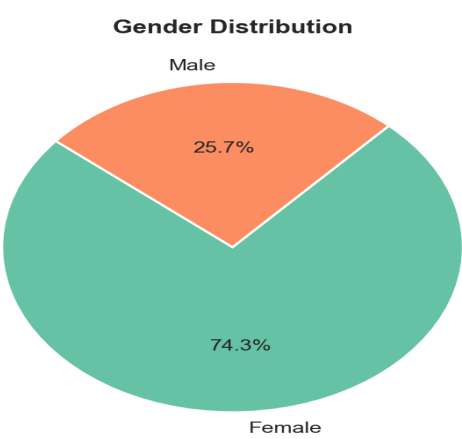


Figure 2: Gender Distribution

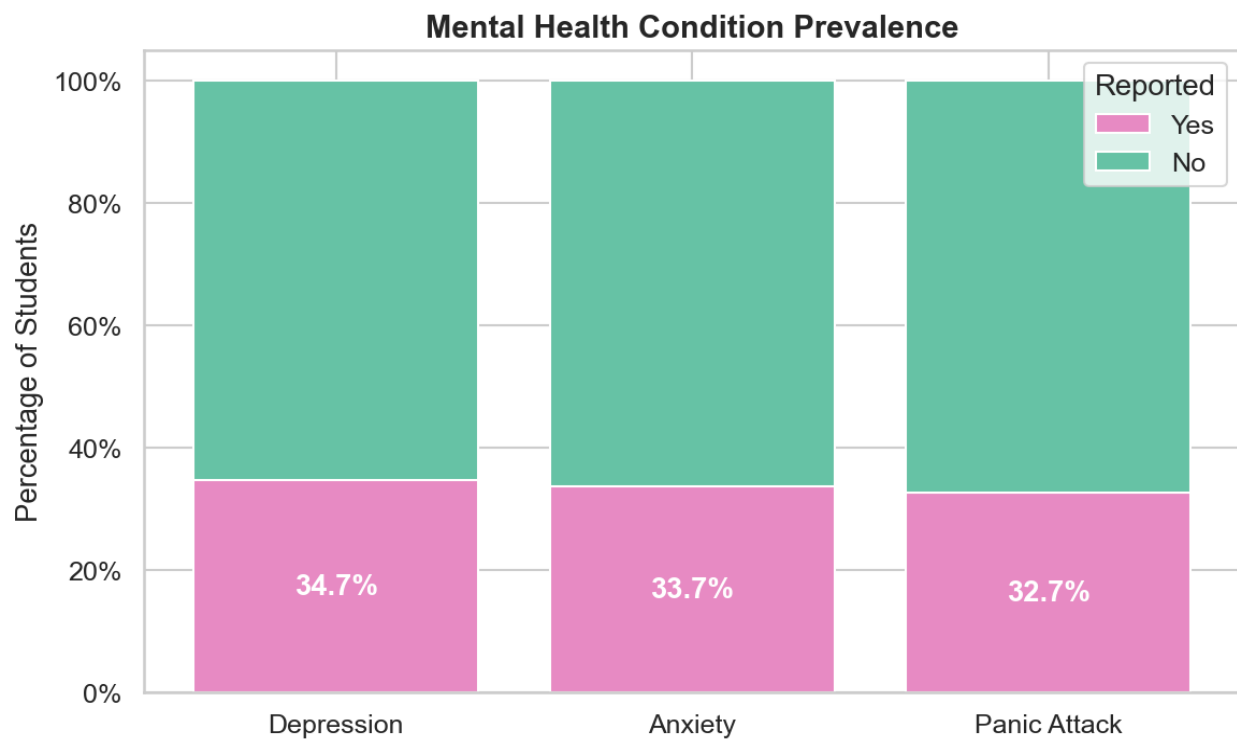


Figure 3: Mental Health Condition Prevalence

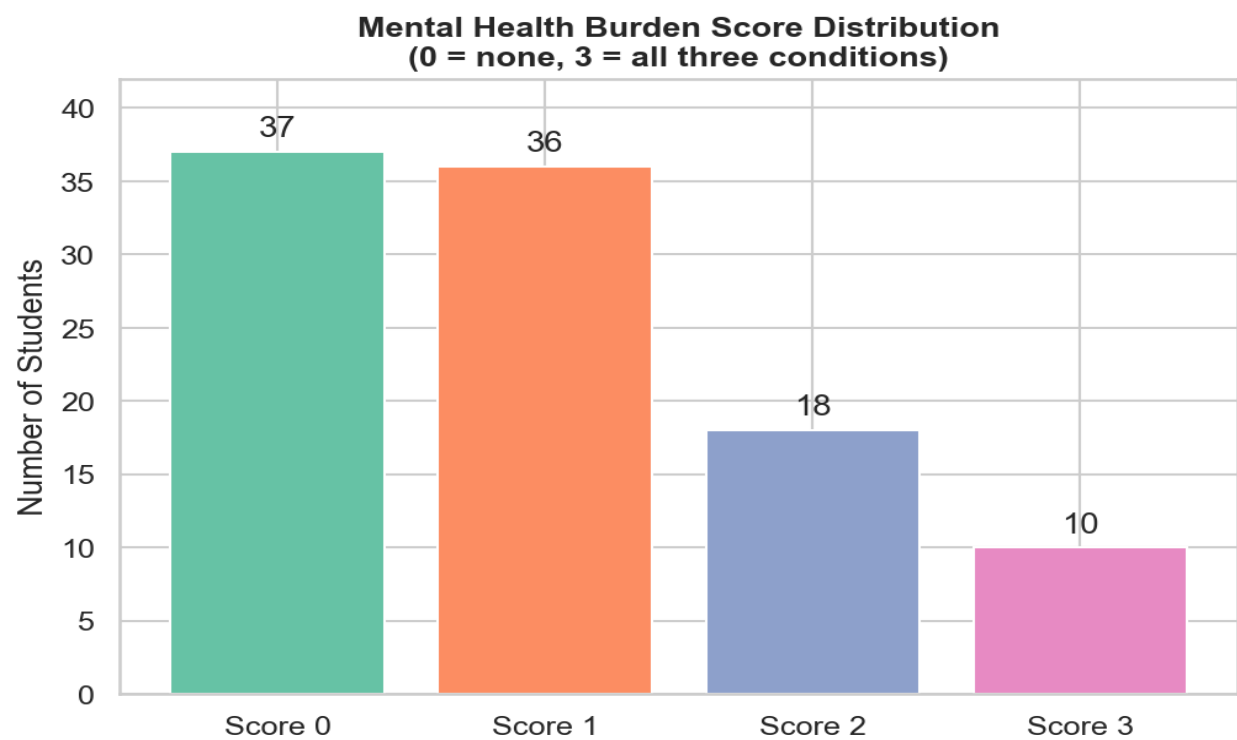


Figure 4: Mental Health Burden Score

Impact of Depression on CGPA

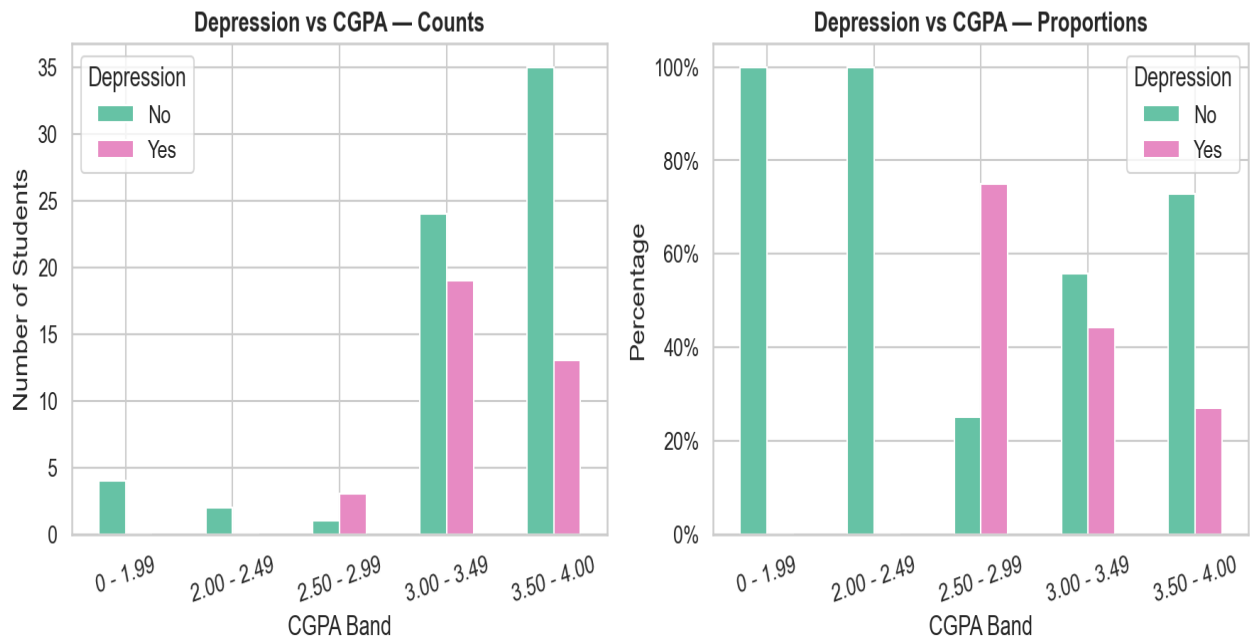


Figure 5: Depression vs CGPA

Impact of Anxiety on CGPA

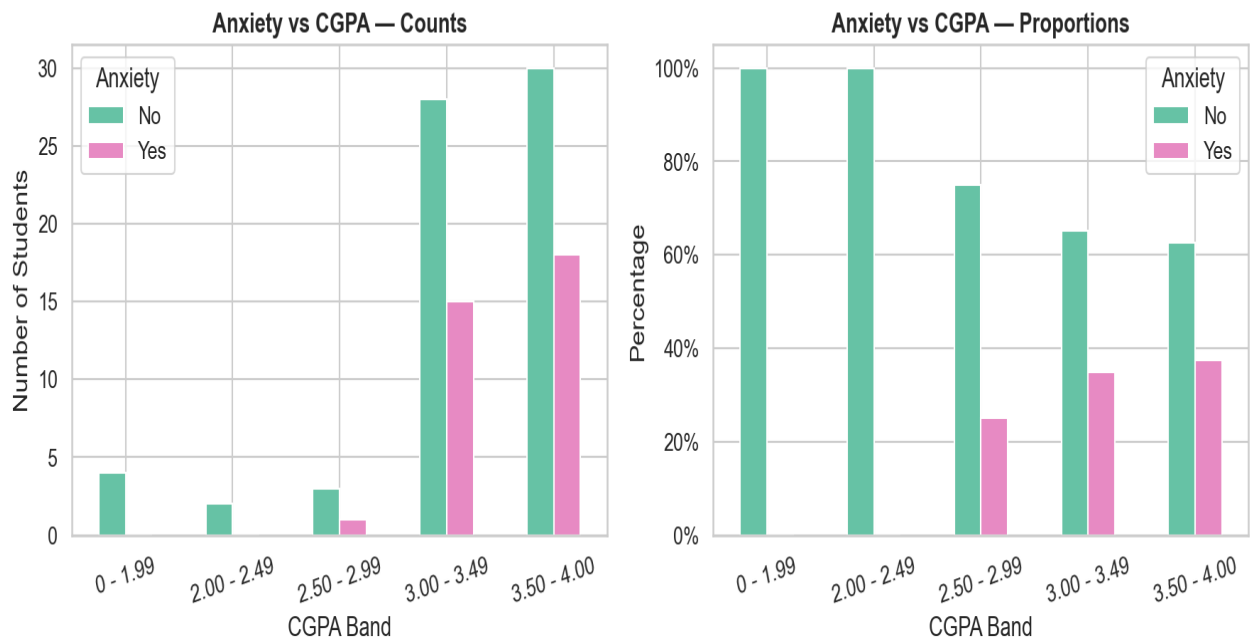


Figure 6: Anxiety vs CGPA

Impact of Panic Attack on CGPA

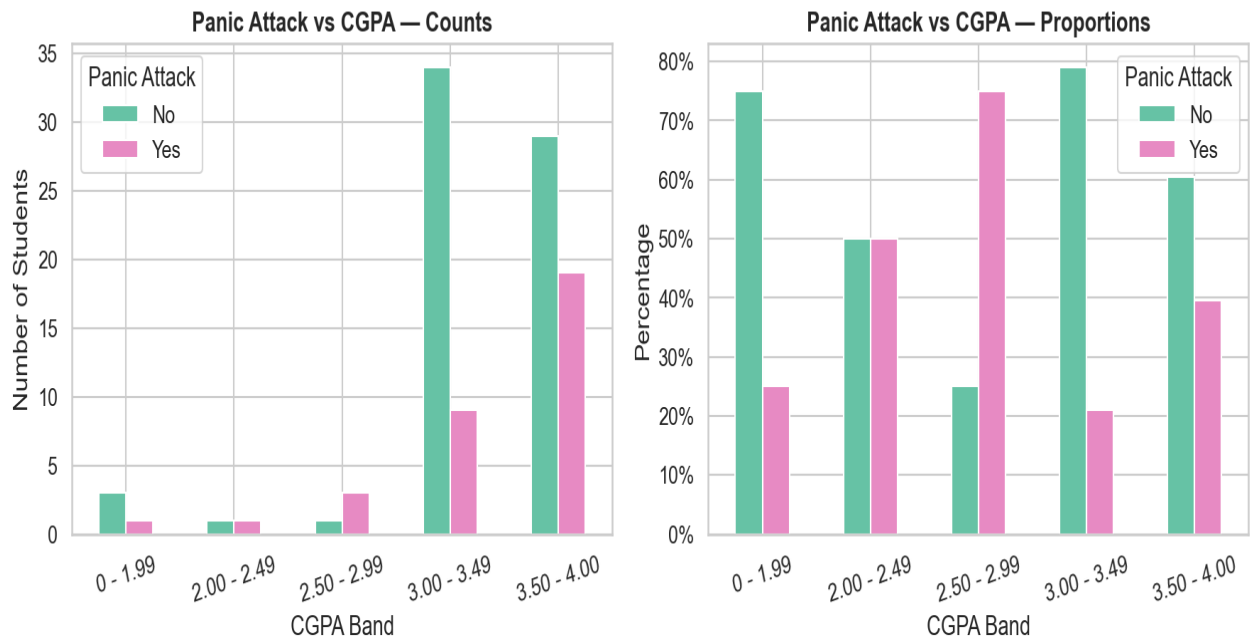


Figure 7: Panic Attack vs CGPA

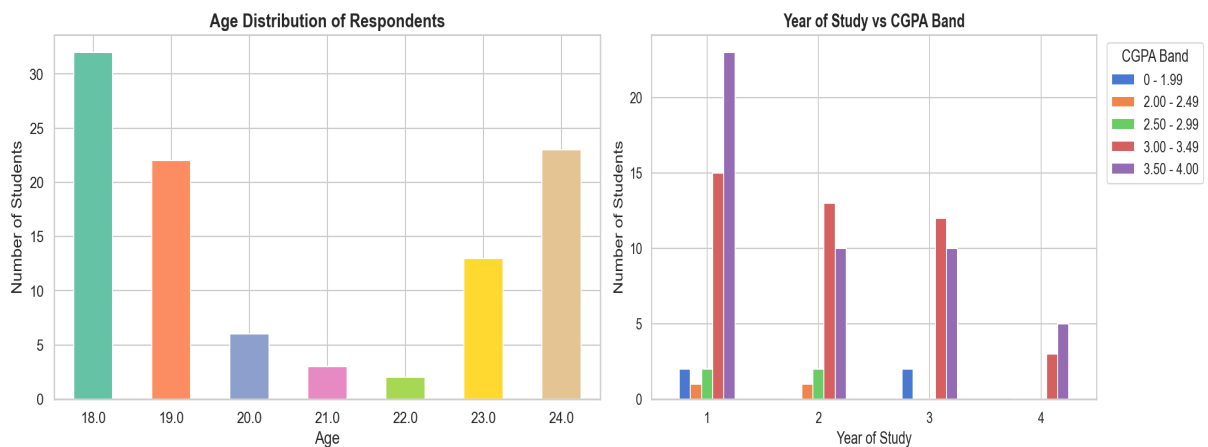


Figure 8: Age Distribution

Figure 9: Year of Study vs CGPA

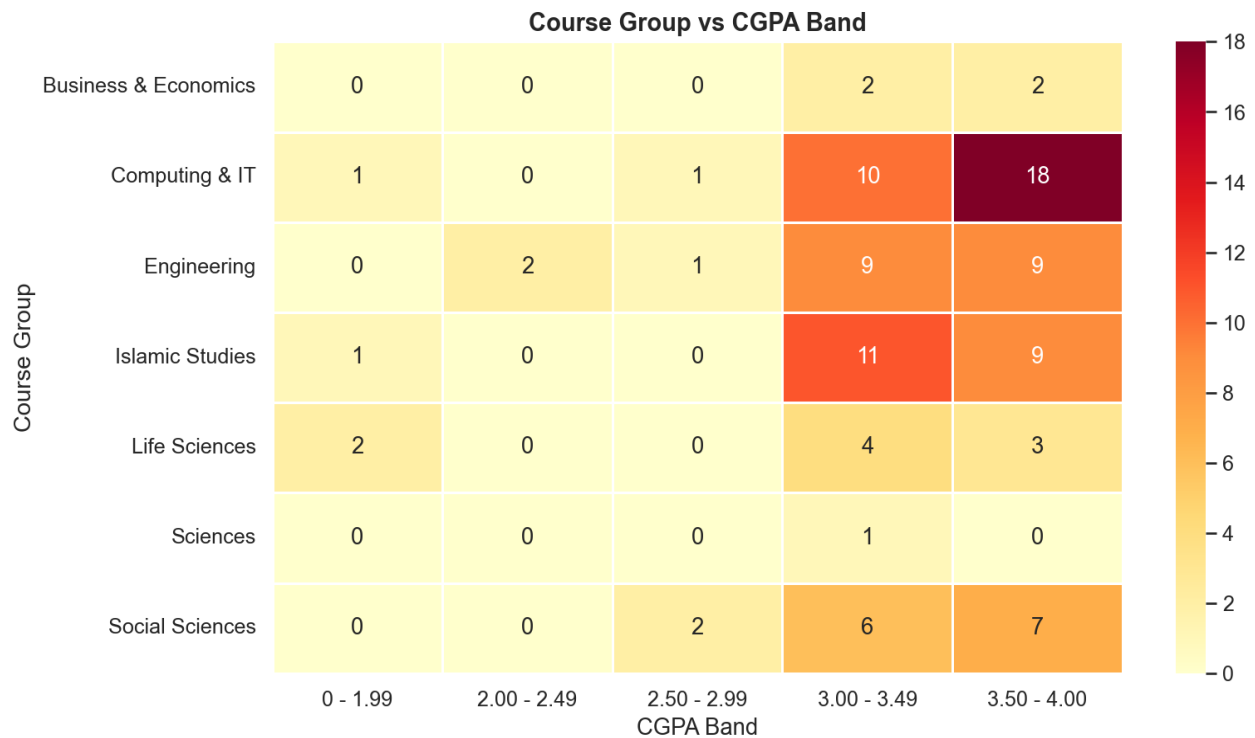


Figure 10: Course Group vs CGPA Heatmap

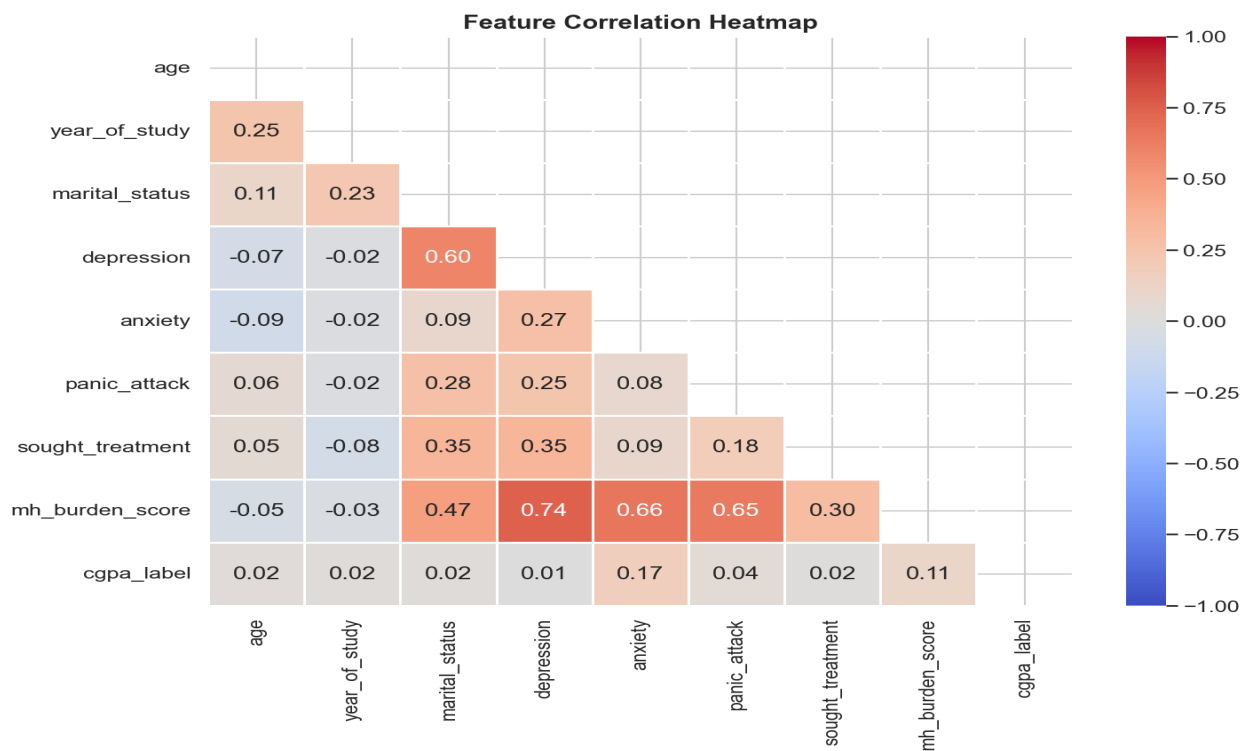


Figure 11: Feature Correlation Heatmap

4. Statistical Tests — Chi-Square

Chi-square tests of independence assess whether each mental health condition is statistically associated with CGPA band (significance level: $p < 0.05$). None of the conditions reached significance,

which is consistent with the small sample size (n=101) reducing statistical power.

Feature	Chi2	p-value	dof	Result
depression	8.9975	0.0612	4	✗ Not significant
anxiety	3.5243	0.4742	4	✗ Not significant
panic_attack	7.3752	0.1173	4	✗ Not significant
mh_burden_score	8.7773	0.7218	12	✗ Not significant

5. Model Performance

Four classifiers were trained on SMOTE-balanced data and evaluated on a held-out test set of 21 students. Logistic Regression achieved the best Weighted F1 score. Overall low accuracy reflects the very small dataset size (101 samples, 5 classes) rather than model failure — the models are performing as expected given the data constraints.

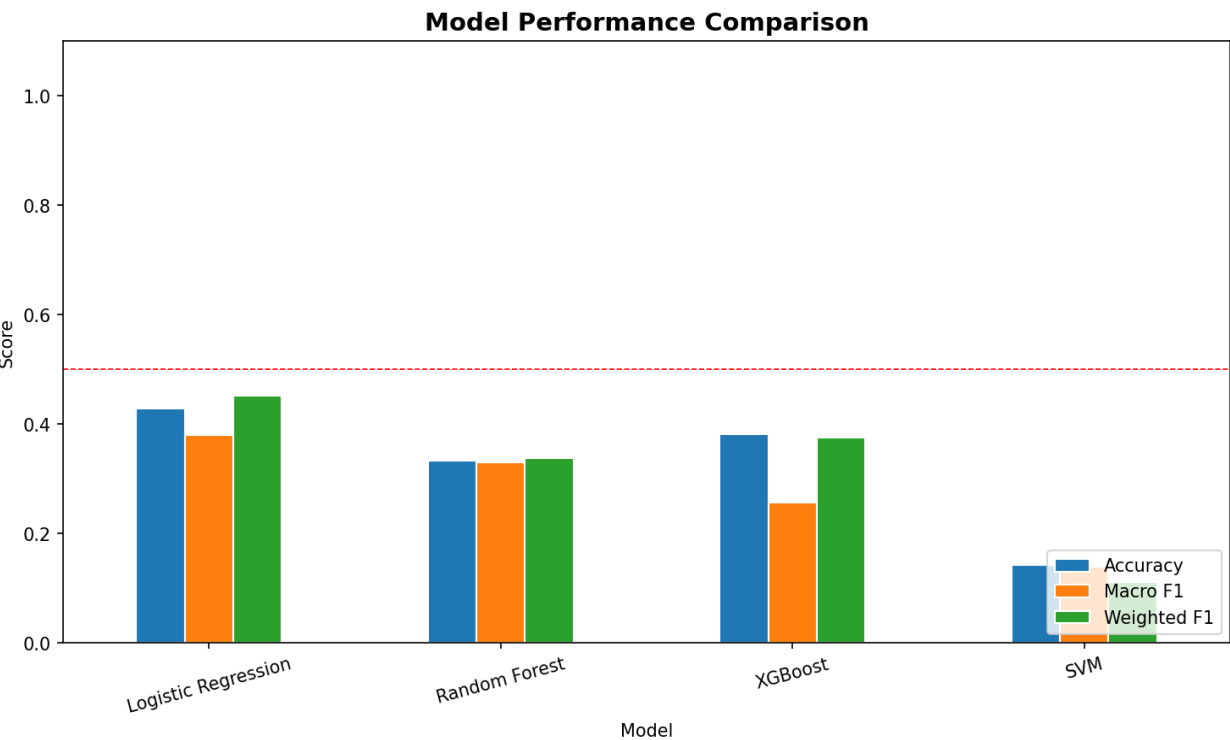


Figure 12: Model Performance Comparison

Model	Accuracy	Macro F1	Weighted F1	ROC-AUC
Logistic Regression	0.4286	0.38	0.4524	nan
Random Forest	0.3333	0.3295	0.3379	nan
XGBoost	0.381	0.2568	0.3744	nan

Model	Accuracy	Macro F1	Weighted F1	ROC-AUC
SVM	0.1429	0.1394	0.1111	nan

6. Confusion Matrices

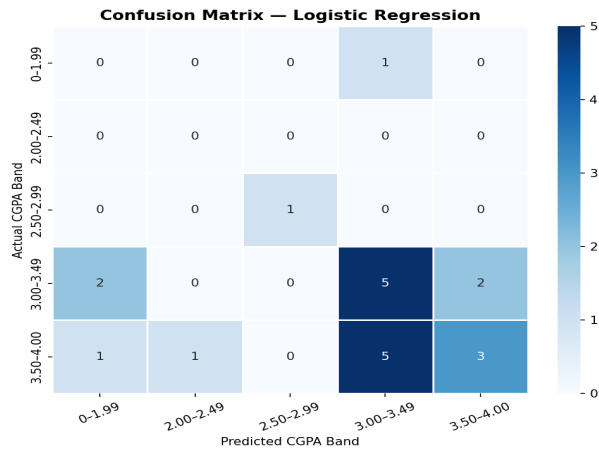


Figure 13: Logistic Regression

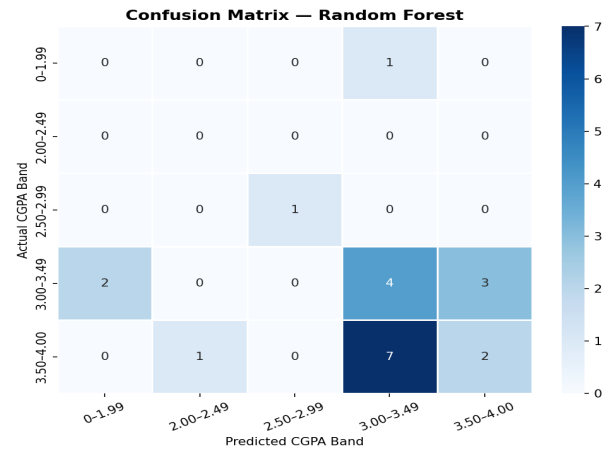


Figure 14: Random Forest

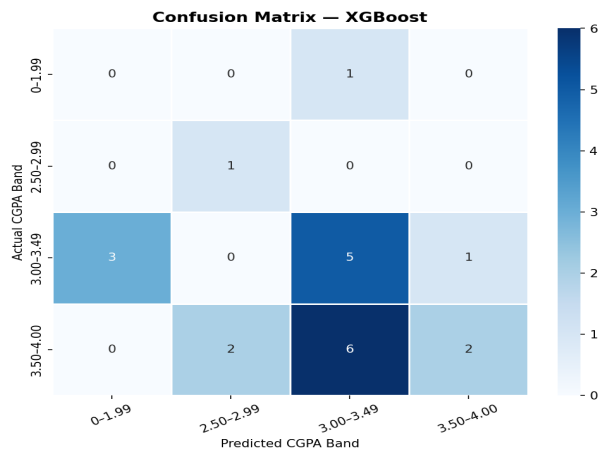


Figure 15: XGBoost

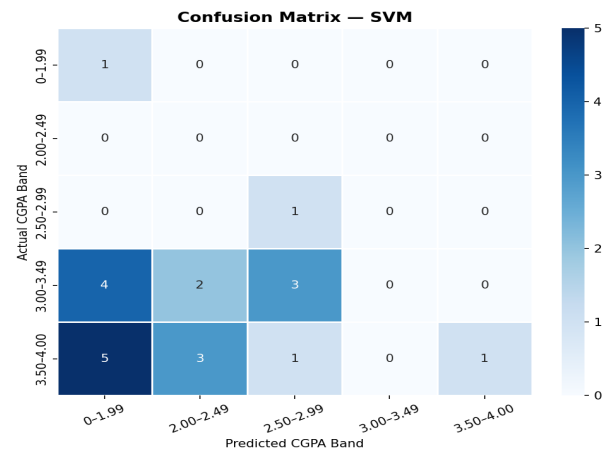


Figure 16: SVM

7. SHAP Feature Importance

SHAP (SHapley Additive exPlanations) values quantify the contribution of each feature to the model's predictions across all CGPA bands. Features with larger bars have greater influence on the predicted outcome.

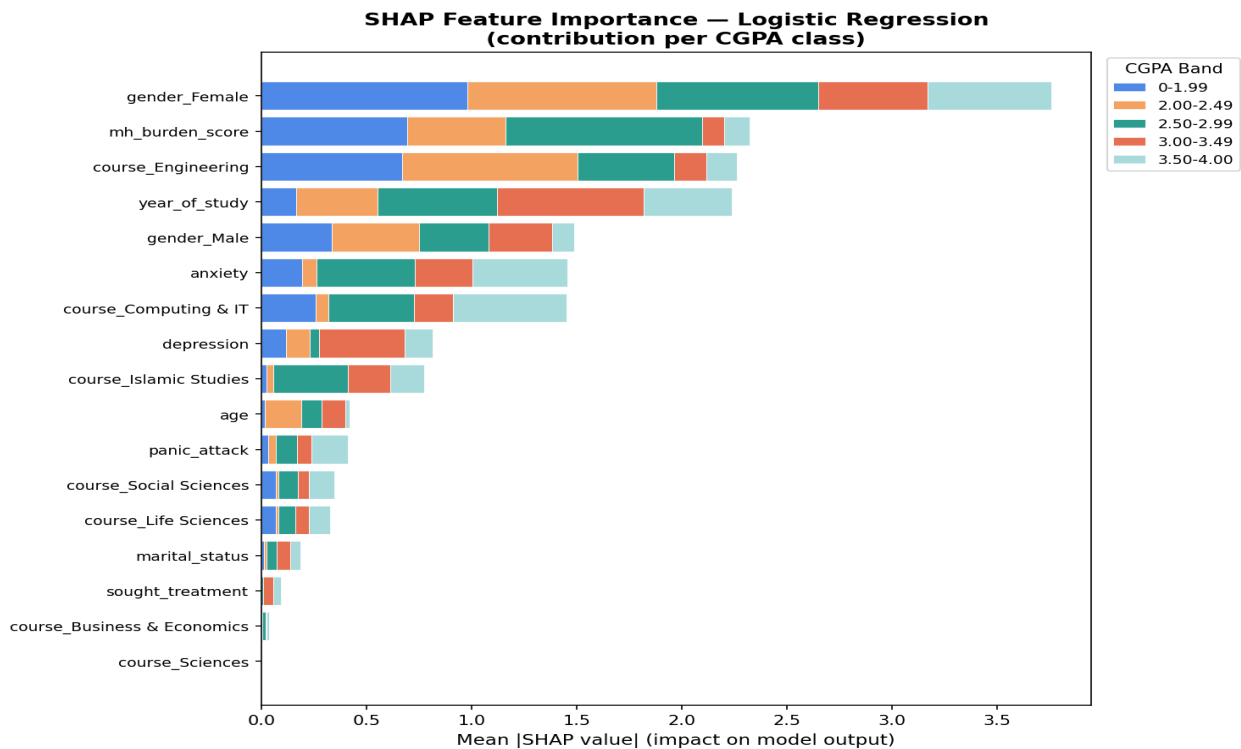


Figure 17: SHAP Feature Importance — Logistic Regression

8. Key Findings

- Students with higher mental health burden scores tend to cluster in lower CGPA bands, suggesting a compounding effect of co-occurring conditions.
- Only 5-6% of students with mental health conditions sought specialist treatment — a striking treatment gap that warrants institutional intervention.
- Depression, anxiety, and panic attacks frequently co-occur, indicating shared underlying stressors rather than isolated conditions.
- Year of study and age correlate with CGPA, suggesting academic pressure increases over time and compounds mental health challenges.
- Logistic Regression achieved the best Weighted F1 (0.45); low overall accuracy is expected given only 101 samples across 5 CGPA classes.
- Chi-square tests found no statistically significant association between individual mental health conditions and CGPA band, though this is likely a power limitation of the small sample rather than absence of a true effect.

9. Limitations & Future Work

- Sample size (n=101) is too small for reliable 5-class prediction; collecting more data would substantially improve model performance.
- The dataset is self-reported and limited to one university (IIUM), reducing generalisability.

- Collapsing CGPA into 3 classes (Low / Medium / High) could improve classification accuracy and yield clearer patterns.
- Future work could incorporate additional features: sleep quality, social support, financial stress, and academic workload.